

The Laplacian Spectrum of the Truncated Octahedron Face Adjacency Graph

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Abstract

We compute the exact spectrum of the graph Laplacian on the face adjacency graph of the truncated octahedron (Kelvin cell). The 14 faces (6 squares, 8 hexagons) form a graph where two faces are adjacent if they share an edge. The eigenvalues are 0 (x1), $(9-\sqrt{17})/2$ (x3), 4 (x2), $(9+\sqrt{17})/2$ (x3), 7 (x4), 9 (x1). The characteristic polynomial factors as $p(\lambda) = \lambda (\lambda^2 - 9\lambda + 16)^3 (\lambda - 4)^2 (\lambda - 7)^4 (\lambda - 9)$. All eigenvalues lie in $\mathbb{Q}(\sqrt{17})$. Each eigenspace is identified with an irreducible representation of the octahedral group O_h .

Keywords: truncated octahedron, Kelvin cell, graph Laplacian, spectral graph theory, octahedral group

1. Introduction

The truncated octahedron is the unique Archimedean solid that tiles $\mathbb{R}^3$. It is the Voronoi cell of the BCC lattice and the solution to Kelvin's problem [1]. We compute the complete Laplacian spectrum of its face adjacency graph and decompose each eigenspace under the full octahedral symmetry group O_h .

2. The Face Adjacency Graph

The truncated octahedron has $F = 14$ faces: 6 squares (normals along $\pm x, \pm y, \pm z$) and 8 regular hexagons (normals along $(\pm 1, \pm 1, \pm 1)/\sqrt{3}$). Two faces are adjacent if they share a polyhedron edge. Square faces have degree 4 (all hexagonal neighbors). Hexagonal faces have degree 6 (3 square + 3 hexagonal neighbors). The face graph has 36 edges. No two squares are adjacent.

3. Spectrum

The graph Laplacian $L = D - A$ has eigenvalues:

Eigenvalue	Exact value	Mult.
λ_1	0	1
λ_2	$(9-\sqrt{17})/2$	3
λ_3	4	2
λ_4	$(9+\sqrt{17})/2$	3
λ_5	7	4
λ_6	9	1

The irrational pair are roots of $x^2 - 9x + 16 = 0$. Sum = 9 (the maximum eigenvalue). Product = 16 = 2^4 . Discriminant = 17 (prime).

Characteristic polynomial: (1)

$$p(\lambda) = \lambda (\lambda^2 - 9\lambda + 16)^3 (\lambda - 4)^2 (\lambda - 7)^4 (\lambda - 9)$$

Verified numerically: $\det(L - 3.5I) = 24790.94$ from both matrix and formula.

4. Trace Identities

$\text{Tr}(L) = 0 + 27 + 8 + 28 + 9 = 72 = 6(4)+8(6)$. The $\sqrt{17}$ terms cancel: $3(9-\sqrt{17})/2 + 3(9+\sqrt{17})/2 = 27$.

$\text{Tr}(L^2) = 456$. Using $r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1r_2 = 81 - 32 = 49$: $3(49) + 0 + 2(16) + 4(49) + 81 = 456 = \text{Sum}(\deg^2) + 2|E| = 384 + 72$.

5. O_h Symmetry Decomposition

The 14-dimensional face permutation representation of O_h decomposes as $A_{1g}(2) + E_g(1) + T_{2g}(1) + A_{2u}(1) + T_{1u}(2)$. The Laplacian commutes with all O_h permutations, so eigenspaces are unions of irreps. Characters were computed by constructing all 48 group elements and computing traces on eigenspace projectors:

Eigenvalue	Dim	Irrep
0	1	A_{1g}
$(9-\sqrt{17})/2$	3	T_{1u}
4	2	E_g
$(9+\sqrt{17})/2$	3	T_{1u}
7	4	$A_{1g} + T_{2g}$
9	1	A_{2u}

The two T_{1u} copies (each 3-dimensional) correspond to the irrational eigenvalue pair. Their splitting is governed by the quadratic $x^2 - 9x + 16 = 0$. The eigenspace at $\lambda = 7$ carries $A_{1g} + T_{2g}$ ($1+3 = 4$), the only eigenspace with more than one irrep.

6. Reproduction

All results are reproducible by: (1) vertices as permutations of $(0, +/ -1, +/ -2)$, (2) identifying 14 faces by normal directions, (3) building the 14×14 adjacency matrix (faces sharing ≥ 2 vertices), (4) computing $L = D - A$, (5) diagonalising. The computation requires only integer arithmetic and one square root ($\sqrt{17}$).

7. Conclusion

$\text{Spec}(L) = \{0^1, ((9-\sqrt{17})/2)^3, 4^2, ((9+\sqrt{17})/2)^3, 7^4, 9^1\}$. Characteristic polynomial: $p(\lambda) = \lambda(\lambda^2 - 9\lambda + 16)^3(\lambda - 4)^2(\lambda - 7)^4(\lambda - 9)$. All eigenvalues algebraic over $\mathbb{Q}(\sqrt{17})$. Irrep decomposition complete under O_h .

References

- [1] Thomson, W. (1887). On the division of space with minimum partitional area. *Phil. Mag.* 24, 503.
- [2] Coxeter, H. S. M. (1973). *Regular Polytopes* (3rd ed.). Dover.
- [3] Dresselhaus, M. S., Dresselhaus, G., Jorio, A. (2008). *Group Theory: Application to Condensed Matter*. Springer.

AI Disclosure

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